DS607 – Statistics for Data Science | Master of Data Science in Economics (MDSE)

Does Investor Fear Predict Stock Returns?

Evidence from VIX and the S&P 500 (2001–2025)

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MOTIVATION

Why Would Fear Predict Returns?

EMH

Efficient Markets View

Standard finance theory holds that prices already reflect all available information — so investor "mood" should carry no predictive power over future returns.

BF

Behavioral Finance View

Irrational sentiment — waves of excessive optimism or pessimism — can persist and move prices away from fundamentals, since limits to arbitrage prevent informed traders from correcting mispricing immediately.

Schmeling (2009, Journal of Empirical Finance): consumer confidence across 18 countries → significant **negative** relation with future returns. Since VIX moves **opposite** to confidence, this project predicts the mirror image: a **positive** relation between lagged VIX and subsequent returns.

FRAMING

Research Question

Does the VIX (as a proxy for investor fear/sentiment) predict subsequent S&P 500 returns, and does this relationship differ between high-VIX and low-VIX periods?

1. Is there a statistically significant relationship between lagged VIX and next-month S&P 500 returns?
2. Does adding dividend yield as a control change that relationship?
3. Does the strength of this relationship differ between calm and turbulent (e.g. COVID-era) periods?
4. Are standard OLS confidence intervals reliable, given VIX's persistence and non-normal returns?

SECTION 1

Data Sources

Variable	Description	Source	Frequency	Coverage
S&P 500 return	MthPrcRet, MthTotRet — monthly price & total return	WRDS — CRSP "Monthly Index & Portfolios on S&P 500"	Monthly	Jan 2001–Dec 2025
VIX	CBOE Volatility Index, daily close	WRDS (CBOE / index database)	Daily → month-end	Jan 2001–Dec 2025
Dividend yield	Constructed as MthTotRet − MthPrcRet	Derived from CRSP file (no separate source)	Monthly	Jan 2001–Dec 2025

The **VIX (CBOE Volatility Index)** is published by the Chicago Board Options Exchange (CBOE). It is calculated from S&P 500 option prices and reflects the market's expectation of volatility over the next 30 days.

⚠ **Data access:**

Raw CRSP / CBOE VIX data are sourced via WRDS institutional access and cannot be redistributed in the public repository (data/raw/ is not tracked).

299

monthly observations in the final merged analysis dataset

METHODOLOGY

Seven-Step Analytical Pipeline

1. **Data preparation** — Clean & merge; construct dividend yield; resample VIX to month-end; lag VIX/DIV one month vs. next-month return

2. **Descriptive analysis** — Time series & scatter plots; histogram/Q-Q + Shapiro-Wilk; VIF multicollinearity check

3. **Baseline regression** — $R(t+1) = \alpha + \beta \cdot VIX(t) + \varepsilon$, OLS, $H_o: \beta = 0$

4. **Extended regression** — + dividend yield control; OLS & HAC (Newey-West) SEs; joint F-test

5. **Block-bootstrap inference** — Block length ≈ n^(1/3), ~ 7 months; 5,000 replications; preserves time-series dependence

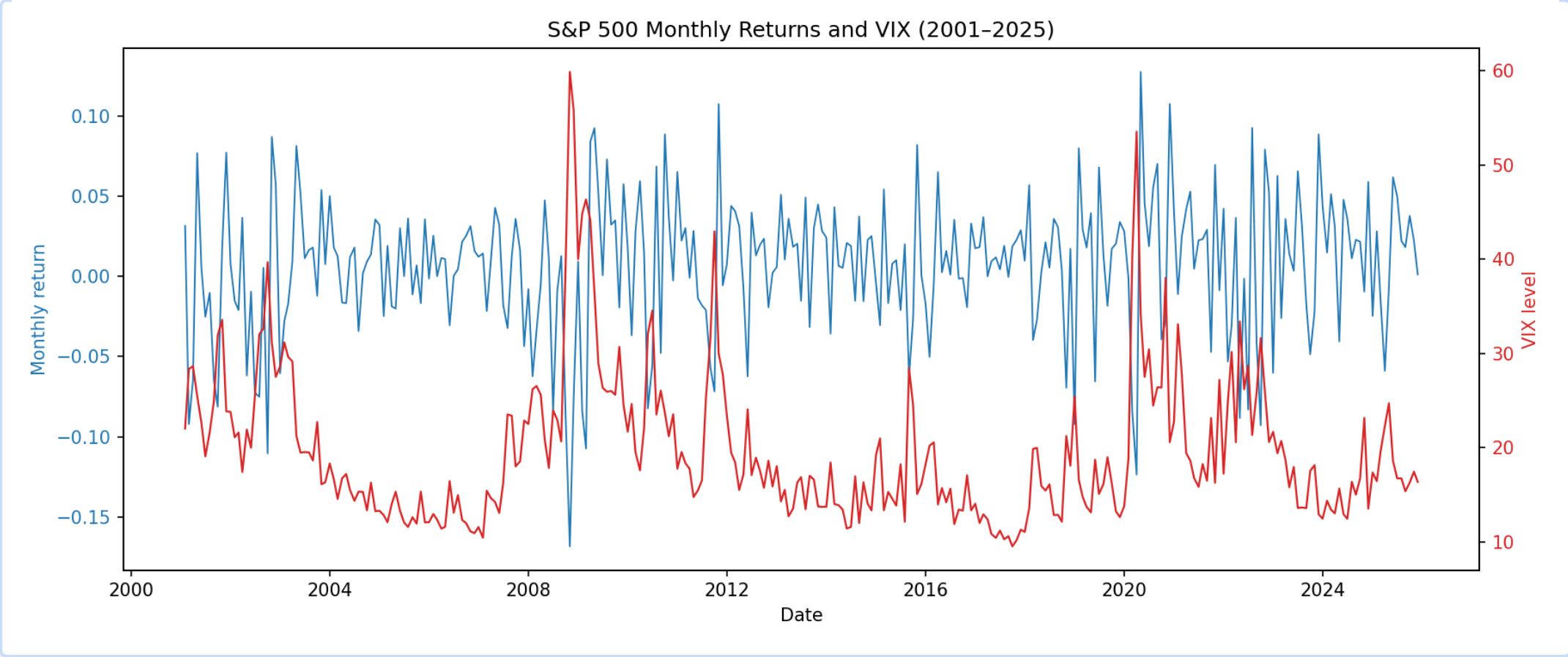
6. **Regime comparison** — High- vs. low-VIX: Welch's t-test (unequal variances)

7. **Permutation test** — 5,000 shuffles of next-month return; assumption-free null for β

5 independent methods converge on the same answer

DESCRIPTIVE OVERVIEW

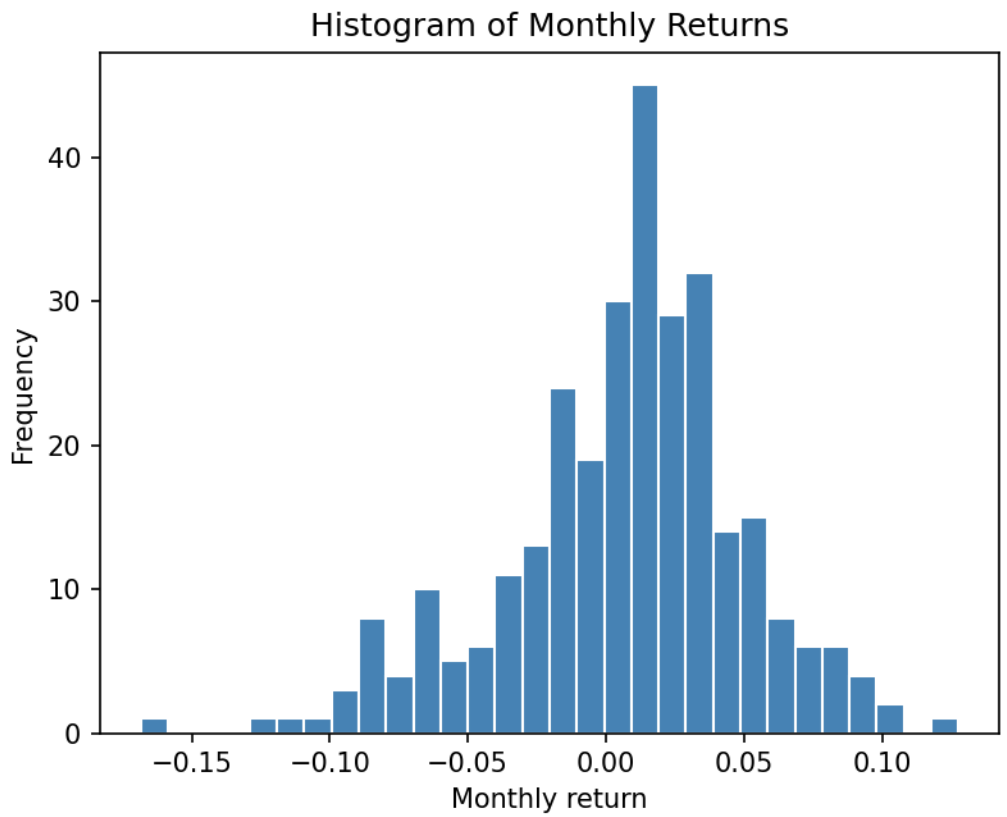
VIX and S&P 500 Returns, 2001–2025

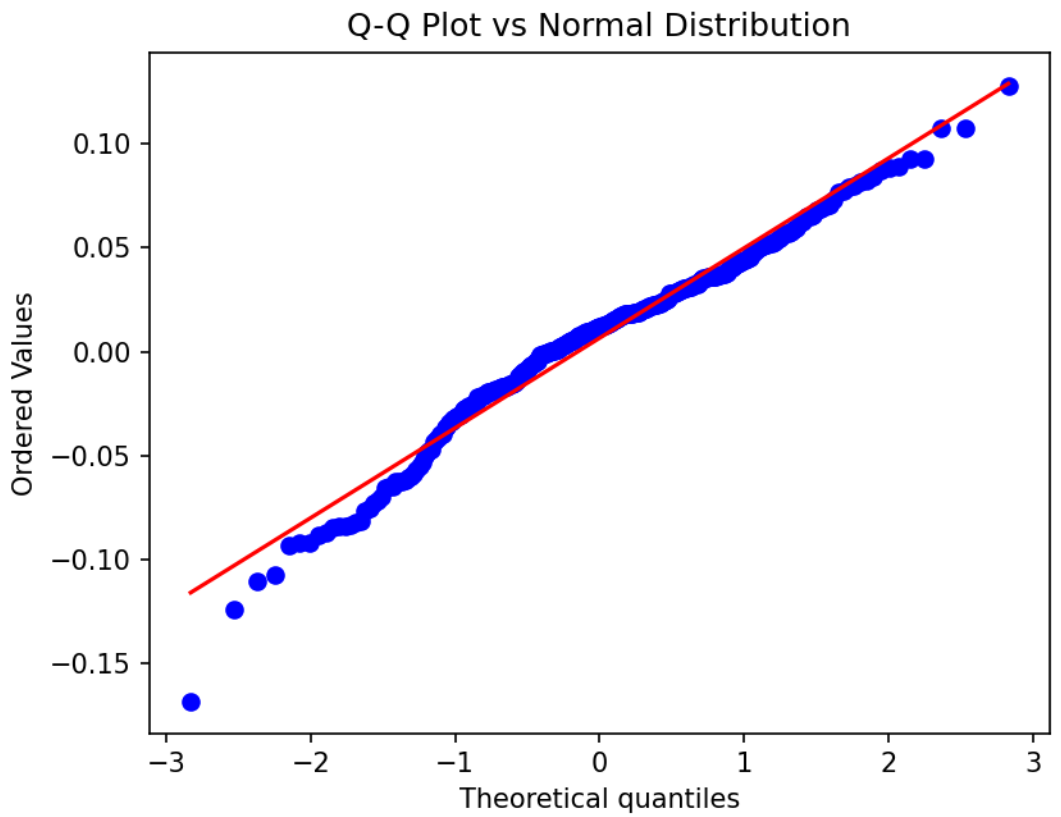


GFC (2008) and COVID (2020) spikes stand out clearly — giving the high/low-VIX regime split meaningful coverage of turbulent periods beyond COVID alone.

DESCRIPTIVE OVERVIEW

S&P 500 Returns Distribution





The histogram looks roughly bell-shaped, but the Q-Q plot shows that the returns have fatter tails than a normal distribution, especially on the downside. The extreme crash months are the main reason. That is why we check the results with methods that don't rely on normality.

SECTION 3–4

Regression Results: Baseline vs. Extended

Table A — Baseline: $R(t+1) = \alpha + \beta \cdot VIX(t) + \varepsilon$ ($R^2 = 0.0012$, Adj. $R^2 = -0.0022$)

Coefficient	Estimate	Std. Error	t-stat	p-value
α (Intercept)	0.00268	0.00676	0.396	0.692
β (VIX)	0.000191	0.000318	0.599	**0.549**

Table B — Extended: $R(t+1) = \alpha + \beta 1 \cdot VIX(t) + \beta 2 \cdot DIV(t) + \varepsilon$ ($R^2 = 0.0013$, Adj. $R^2 = -0.0054$)

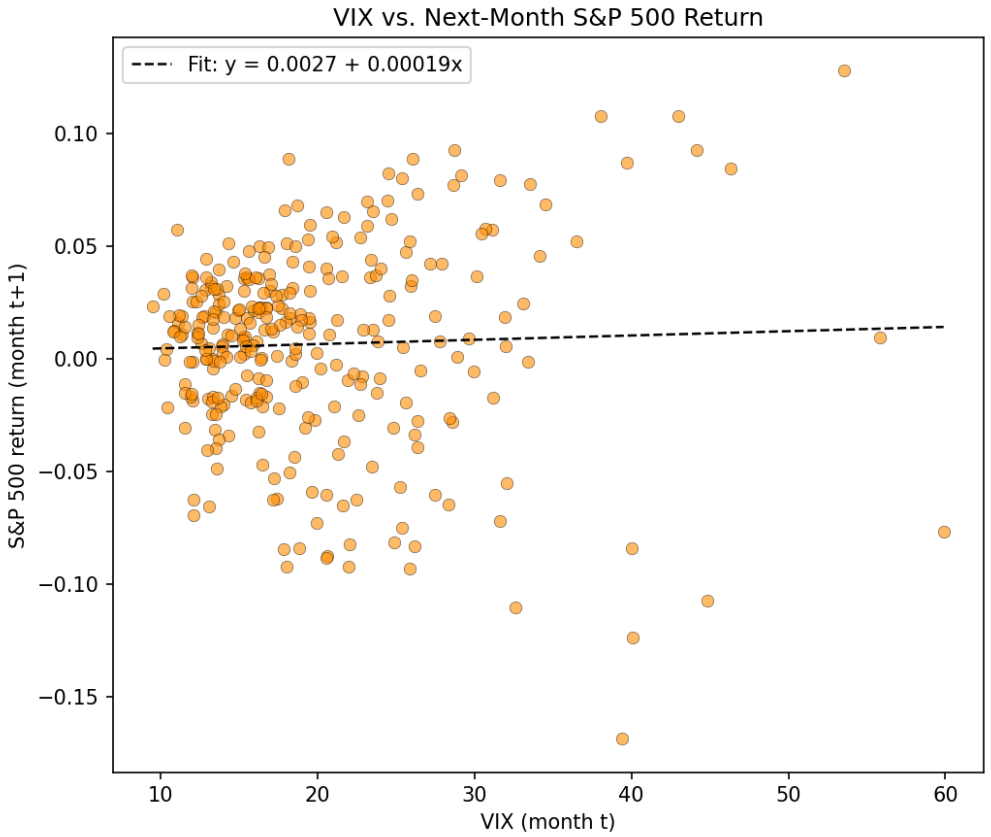
Coefficient	Estimate	SE (OLS)	SE (HAC)	t (OLS)	p (OLS)
α (Intercept)	0.00129	0.00929	0.01177	0.139	0.890
β1 (VIX)	0.000185	0.000320	0.000562	0.579	0.563
β2 (DIV)	0.935	4.282	4.064	0.218	0.827

Baseline: VIX p = 0.549, fail to reject H_o (the claim that VIX has no effect (H_o: $\beta = 0$))
Extended: VIX p = 0.563 and dividend yield p = 0.827 individually (H_o: $\beta_1 = 0$ and H_o: $\beta_2 = 0$), joint F-test p = 0.817 (H_o: $\beta_1 = \beta_2 = 0$).

Practical Implications:
We fail to reject the null for VIX and dividend yield individually, and for both jointly (F-test p = 0.817). This is consistent with R^2 of 0.0012, which means VIX explains only about 0.12% of the variation in monthly returns, so returns are driven mostly by factors it does not capture.
Adding dividend yield lowers the adjusted R^2, so it does not improve the model enough to justify including it.

VISUAL CHECK

VIX vs. Next-Month Return



What the scatter shows

A mild **positive slope** — consistent with the mirror-image-of-Schmeling prediction (higher fear → slightly higher next-month return).

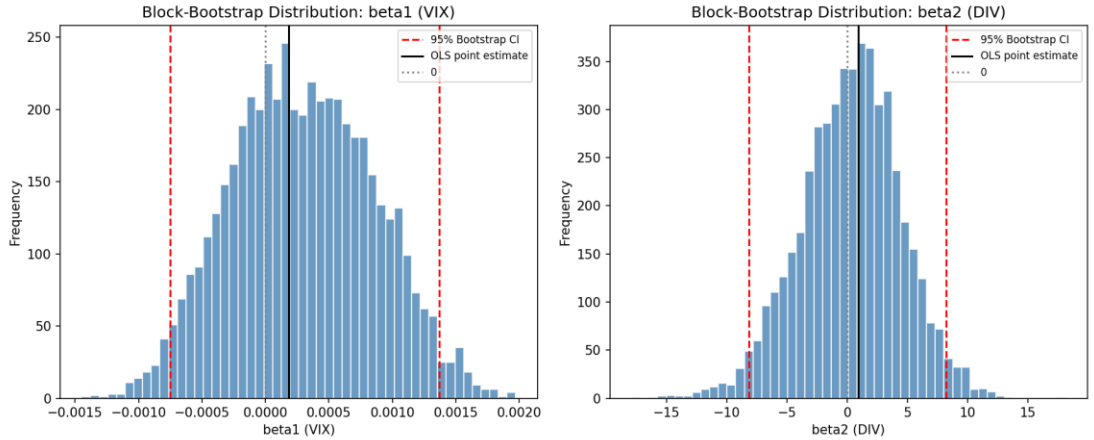
But: the slope is not statistically significant (see baseline regression, Slide 8: p = 0.549).

WHY STANDARD SEs MIGHT NOT BE ENOUGH

Robustness Check 1: Block Bootstrap

VIX is highly persistent, so an i.i.d. bootstrap would destroy time-series dependence. A **moving block bootstrap** (block length ≈ n^(1/3), 5,000 replications) resamples whole rows to preserve that structure.

Coefficient	Method	95% CI Lower	95% CI Upper	Contains 0?
β1 (VIX)	OLS	−0.00044	0.00081	Yes
β1 (VIX)	HAC	−0.00092	0.00129	Yes
β1 (VIX)	Block Bootstrap	−0.00075	0.00137	Yes
β2 (DIV)	OLS	−7.49	9.36	Yes
β2 (DIV)	HAC	−7.03	8.90	Yes
β2 (DIV)	Block Bootstrap	−8.16	8.25	Yes



Bootstrap sampling distributions

All three 95% CIs for β1 and β2 contain zero — block bootstrap agrees with OLS and HAC.

Practical Implications:

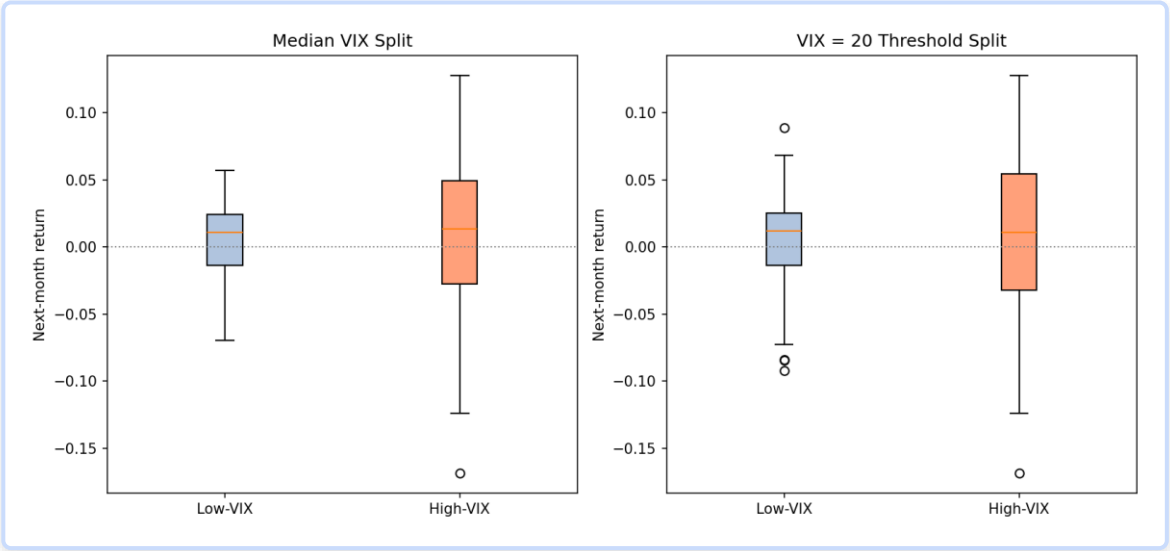
Every 95% interval for VIX and dividend yield contains zero, under OLS, HAC and the block bootstrap. The data are consistent with no effect, so we find no reliable evidence that either variable predicts returns. The data also cannot rule out small effects in either direction.

WELCH'S T-TEST

Robustness Check 2: High- vs. Low-VIX Regimes

Split	N high	N low	Mean high	Mean low
Median VIX	150	149	0.00651	0.00637
VIX = 20	107	192	0.00682	0.00623

Split	N high	N low	t-stat	p-value
Median VIX	150	149	0.028	0.978
VIX = 20	107	192	0.095	0.924



Means: statistically indistinguishable

Welch's t-test finds no significant mean difference across either split (p = 0.978 / p = 0.924).

Dispersion: visibly wider in high-VIX months

SD ≈ 0.056–0.060 (high) vs. ≈ 0.026–0.031 (low) — roughly double the spread.

Practical Implications:

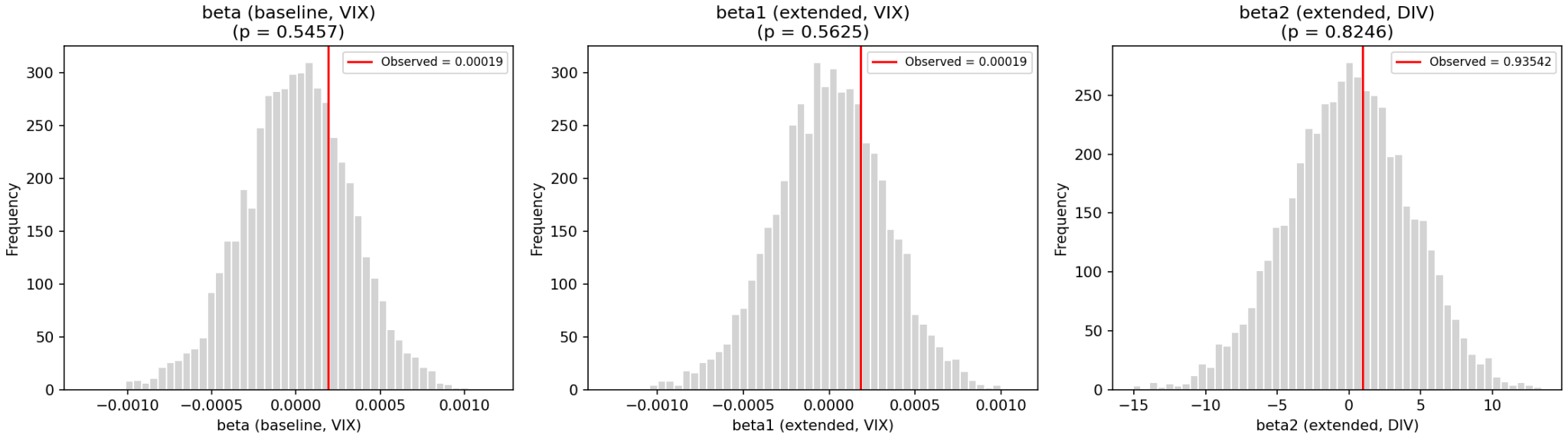
The boxplot shows that returns in high-VIX months are more spread out, with a larger box and longer whiskers, while the average return is about the same. The spread is wider in both directions, so VIX tells us about the size of market moves rather than their direction.

ASSUMPTION-FREE NULL

Robustness Check 3: Permutation Test

Question:

if VIX had no relationship with next-month returns, how often would a coefficient as large as the one observed appear by chance?



Coefficient	Observed Estimate	Permutation p	OLS p
β (baseline, VIX)	0.000191	0.546	0.549
β1 (extended, VIX)	0.000185	0.562	0.563
β2 (extended, DIV)	0.935	0.825	0.827

Non-parametric test using permutation p-values closely track OLS p-values — reinforcing non-significance from a normality assumption angle.

KEY FINDING

Across **5 independent inference methods** (OLS, HAC, block bootstrap, Welch's t-test, permutation), VIX does **NOT** show a statistically significant predictive relationship with next-month S&P 500 returns over 2001–2025 — alone or with dividend yield as a control.

SECONDARY FINDING

VIX does not predict return **DIRECTION** — but high-VIX months show visibly greater return **DISPERSION.**

VIX may better capture next-month ***volatility*** than next-month ***direction***.

SECTION 8

Limitations

1 Predictive, not causal

VIX is not randomly assigned — relationships found are associational, not causal.

2 COVID as a structural break

March–April 2020's fear spike coincides with an unprecedented policy response that may confound the result.

3 VIX as an imperfect sentiment proxy

Mechanically a volatility measure — any effect may reflect a volatility risk premium, not pure behavioral overreaction.

4 Remaining omitted variables

Lagged market returns, realized volatility, interest-rate spreads — flagged as future work.

CONCLUSION

Investor fear (VIX) shows no significant power to predict the ***direction*** of next-month S&P 500 returns across five independent methods — but may help explain next-month ***volatility*** instead.

Thank you

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